We have $P(E \cap F) = \frac{1}{4}$

$$P(E) = \frac{2}{4}, \quad P(F) = \frac{2}{4}$$

$$P(E)\,P(F) = \frac{2}{4}.\frac{2}{4} = \frac{1}{4}$$

$$\Rightarrow P(E \cap F) = P(E)\,.\,P(F)$$

Therefore E and F are independent events.

Note 12.7

Independent events is a property of probability but mutual exclusiveness is a set-theoretic property. Therefore independent events can be identified by their probabilities and mutually exclusive events can be identified by their events.

Theorem 12.9

Suppose A and B are two events, such that $P(A) \neq 0$, $P(B) \neq 0$.

(1) If A and B are mutually exclusive, they cannot be independent.

(2) If A and B are independent they cannot be mutually exclusive. (Without proof)

Example 12.20

If A and B are two independent events such that $P(A) = 0.4$ and $P(A \cup B) = 0.9$. Find $P(B)$.

Solution

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A).P(B) \qquad \text{(since } A \text{ and } B \text{ are independent)}$$

That is, $0.9 = 0.4 + P(B) - (0.4)\,P(B)$

$$0.9 - 0.4 = (1 - 0.4)\,P(B)$$

Therefore, $P(B) = \frac{5}{6}$.

Example 12.21

An anti-aircraft gun can take a maximum of four shots at an enemy plane moving away from it. The probability of hitting the plane in the first, second, third, and fourth shot are respectively 0.2, 0.4, 0.2 and 0.1. Find the probability that the gun hits the plane.



Solution

Let H_1, H_2, H_3 and H_4 be the events of hitting the plane by the anti-aircraft gun in the first second, third and fourth shot respectively.

Let H be the event that anti-aircraft gun hits the plane. Therefore $\bar{H}$ is the event that the plane is not shot down. Given that

$$P(H_1) = 0.2 \quad \Rightarrow P(\bar{H}_1) = 1 - P(H_1) = 0.8$$

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